

MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 03: Tuesday 18 August 2026

(Week 2)

Remark. See the announcement in canvas about the Math Clinic.



Some Standard Integrals

(See also Thomas' Calculus, A Brief Table of Integrals, T-1, near the end of the book.)

$$1. \int (ax + b)^n dx = \frac{(ax + b)^{n+1}}{(n+1)a} + C \quad (n \neq -1)$$

$$2. \int \frac{1}{ax + b} dx = \frac{1}{a} \ln |ax + b| + C$$

$$3. \int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$4. \int \sin(ax + b) dx = -\frac{1}{a} \cos(ax + b) + C$$

$$5. \int \cos(ax + b) dx = \frac{1}{a} \sin(ax + b) + C$$

$$6. \int \tan(ax + b) dx = \frac{1}{a} \ln |\sec(ax + b)| + C$$

$$\begin{aligned}
 7. \int \sec(ax + b) dx \\
 = \frac{1}{a} \ln |\sec(ax + b) + \tan(ax + b)| + C
 \end{aligned}$$

$$\begin{aligned}
 8. \int \csc(ax + b) dx \\
 = -\frac{1}{a} \ln |\csc(ax + b) + \cot(ax + b)| + C
 \end{aligned}$$

$$9. \int \cot(ax + b) dx = -\frac{1}{a} \ln |\csc(ax + b)| + C$$

$$10. \int \sec^2(ax + b) dx = \frac{1}{a} \tan(ax + b) + C$$

$$11. \int \csc^2(ax + b) dx = -\frac{1}{a} \cot(ax + b) + C$$

$$12. \int \sec(ax + b) \tan(ax + b) dx = \frac{1}{a} \sec(ax + b) + C$$

$$13. \int \csc(ax+b) \cot(ax+b) dx = -\frac{1}{a} \csc(ax+b) + C$$

$$14. \int \frac{1}{a^2 + (x+b)^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x+b}{a}\right) + C$$

$$15. \int \frac{1}{\sqrt{a^2 - (x+b)^2}} dx = \sin^{-1}\left(\frac{x+b}{a}\right) + C$$

$$16. \int \frac{-1}{\sqrt{a^2 - (x+b)^2}} dx = \cos^{-1}\left(\frac{x+b}{a}\right) + C$$

$$17. \int \frac{1}{a^2 - (x+b)^2} dx = \frac{1}{2a} \ln \left| \frac{x+b+a}{x+b-a} \right| + C$$

$$18. \int \frac{1}{(x+b)^2 - a^2} dx = \frac{1}{2a} \ln \left| \frac{x+b-a}{x+b+a} \right| + C$$

$$19. \int \frac{1}{\sqrt{(x+b)^2 + a^2}} dx$$

$$= \ln \left| (x+b) + \sqrt{(x+b)^2 + a^2} \right| + C$$

$$20. \int \frac{1}{\sqrt{(x+b)^2 - a^2}} dx$$

$$= \ln \left| (x+b) + \sqrt{(x+b)^2 - a^2} \right| + C$$

$$21. \int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$$

$$22. \int \sqrt{x^2 - a^2} dx$$

$$= \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \ln |x + \sqrt{x^2 - a^2}| + C$$

Q02.14. Find $\int x \cos(x^2 + 1) dx$.

Q02.15. Find $\int \frac{e^{3x}}{\sqrt{2e^{3x} + 5}} dx$.

Q02.16. Find $\int x \ln x dx$.

Integration by Substitution.

Let $u = g(x)$. Then $\frac{du}{dx} = g'(x)$.

Now think of $\underline{g'(x) dx} = du$. Then

$$\int f(g(x)) \underline{g'(x) dx} = \int f(u) \underline{du}.$$

Remark. Integration by substitution is a consequence of the chain rule in differentiation.

Q02.14. Find $\int x \cos(x^2 + 1) dx$.

Put $u = x^2 + 1$

Q02.14. Find $\int x \cos(x^2 + 1) dx$.

Solution: Put $u = x^2 + 1$. Then $\frac{du}{dx} = 2x$
 $\Rightarrow \frac{1}{2} du = x dx$.

$$\begin{aligned} & \int x \cos(x^2 + 1) dx \\ &= \int \cos u \cdot \frac{1}{2} du \\ &= \frac{1}{2} \sin u + C \\ &= \frac{1}{2} \sin(x^2 + 1) + C \end{aligned}$$

$$\begin{aligned} & \int x \cos(x^2+1) dx \\ &= \int \cos(x^2+1) \cdot \frac{1}{2} d(x^2+1) \\ &= \frac{1}{2} \cdot \sin(x^2+1) + C. \end{aligned}$$

Q02.15. Find $\int \frac{e^{3x}}{\sqrt{2e^{3x} + 5}} dx$.

Solution: Let $u = 2e^{3x} + 5$. Then $\frac{du}{dx} = 6e^{3x}$, so that

$e^{3x} dx = \frac{1}{6} du$. Then

$$\begin{aligned} & \int \frac{e^{3x}}{\sqrt{2e^{3x} + 5}} dx \\ &= \int \frac{1}{6\sqrt{u}} du \\ &= \frac{1}{3} \sqrt{u} + C \\ &= \frac{1}{3} \sqrt{2e^{3x} + 5} + C. \end{aligned}$$

Integration by Parts.

When integrating a product of two functions, try to think of one function as a derivative. Then

or
$$\int \underbrace{f'(x)g(x)} \, dx = f(x) \cdot g(x) - \int \underbrace{f(x)g'(x)} \, dx,$$

$$\int f(x)g'(x) \, dx = f(x) \cdot g(x) - \int f'(x)g(x) \, dx.$$

Remark. The integration by parts is a consequence of the product rule of differentiation.

Q02.16. Find $\int \underline{x \ln x} \, dx$.

$$u = \frac{d}{dx} \left(\frac{x^2}{2} \right)$$

Solution.

$$\begin{aligned} \int x \ln x \, dx &= \frac{x^2}{2} \cdot (\ln x) - \int \frac{x^2}{2} \cdot \frac{1}{x} \, dx \\ &= \frac{x^2 \ln x}{2} - \int \frac{x}{2} \, dx \\ &= \frac{x^2 \ln x}{2} - \frac{x^2}{4} + C. \end{aligned}$$

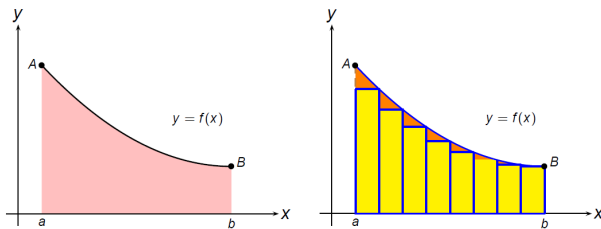
Q02.17. Evaluate $\int_1^e \frac{\ln x}{x} dx$.

Q02.18. Evaluate $\int_0^{\frac{\pi}{2}} x \cos x dx$.

Q02.19. Find $\frac{d}{dx} \int_1^{\sin x} \sqrt{1+t^4} dt$.

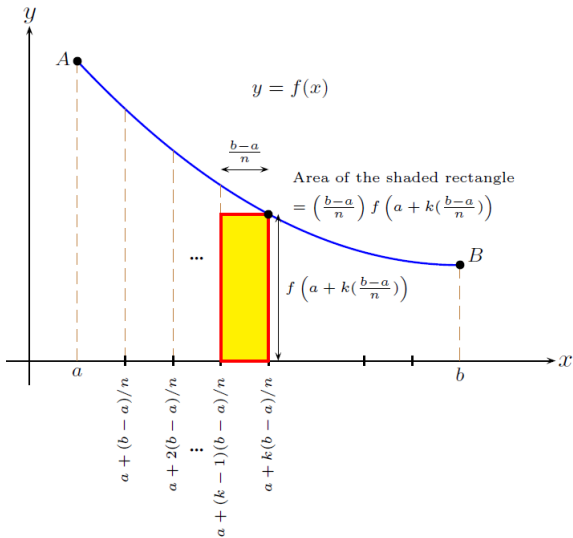
Definite Integral (Section 5.2)

How does one find the area of the region under the graph of a continuous function $f(x)$ from $x = a$ to $x = b$?



$$\int_a^b f(x) dx = \text{area of pink region.}$$

Let f be continuous on $[a, b]$.



Then the following limit exists and is known as the *definite integral* of f from $x = a$ to $x = b$.

$$\lim_{n \rightarrow \infty} \left\{ \sum_{k=1}^n \left(\frac{b-a}{n} \right) f \left(a + k \left(\frac{b-a}{n} \right) \right) \right\}.$$

It is denoted by

$$\int_a^b f(x) dx.$$

Geometrically the definite integral gives the area of the region under the graph of f from $x = a$ to $x = b$ (at least in the case when $f(x) \geq 0$ and $a < b$).

The numbers a and b are respectively called the lower and upper limits of the definite integral. The function $f(x)$ is called the integrand of the definite integral.

The expression $\sum_{k=1}^n \left(\frac{b-a}{n} \right) f \left(a + k \left(\frac{b-a}{n} \right) \right)$ is known as a Riemann sum of f . In summary, we have:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \left\{ \sum_{k=1}^n \left(\frac{b-a}{n} \right) f \left(a + k \left(\frac{b-a}{n} \right) \right) \right\}.$$

Remark. An informal visualization of the above formula:

Think of $f(x)$ dx as the area of a narrow rectangle of height $f(x)$ and width dx .

Also we think of the definite integral sign $\int_a^b$ as the process of 'taking the limit of the (Riemann) sum' (of the areas of these narrow rectangles).

The method of using Riemann sums to evaluate

$\int_a^b f(x) dx$ is tedious. The following result, known as the

Fundamental Theorem of Calculus (FTC Part 2),

gives a simpler way of calculating $\int_a^b f(x) dx$:

Theorem. (FTC Part 2) Let f be continuous on $[a, b]$ and let F be an anti-derivative of f (i.e., $F'(x) = f(x)$). Then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Notation: Often we denote $\left[F(x) \right]_a^b = F(b) - F(a)$.

Then the FTC 1 takes the form

$$\int_a^b f(x) dx = \left[F(x) \right]_a^b = F(b) - F(a).$$

Remark. Often while working out an anti-derivative, we keep the upper and lower limits in our workings.

Q02.17. Evaluate $\int_1^e \frac{\ln x}{x} dx$.

Solution: To find an anti-derivative, we make the substitution $u = \ln x$.

$$\begin{aligned}\int_1^e \frac{\ln x}{x} dx &= \int_{x=1}^{x=e} \ln x d(\ln x) \\ &= \left[\frac{(\ln x)^2}{2} \right]_1^e \\ &= \frac{(\ln e)^2}{2} - \frac{(\ln 1)^2}{2} = \frac{1}{2} - 0 = \frac{1}{2}.\end{aligned}$$

Q02.18. Evaluate $\int_0^{\frac{\pi}{2}} x \cos x \, dx$.

$$\cos x = \frac{d}{dx}(\sin x)$$

Solution: Using integration by parts, we have

$$\begin{aligned}\int_0^{\frac{\pi}{2}} x \cos x \, dx &= \left[x \sin x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x \, dx \\&= \left[x \sin x \right]_0^{\frac{\pi}{2}} + \left[\cos x \right]_0^{\frac{\pi}{2}} \\&= \left(\frac{\pi}{2} - 0 \right) + (0 - 1) \\&= \frac{\pi}{2} - 1.\end{aligned}$$

Next we have the **Fundamental Theorem of Calculus (Part 1) (FTC Part 1)** as follows:

Theorem. (FTC Part 1) Let f be continuous on $[a, b]$.

Then

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$

for all $x \in [a, b]$.

Remark. FTC Part 1 says that the derivative of the definite integral $\int_a^x f(t) dt$ with respect to the upper limit x is equal to the integrand evaluated at the upper limit x . Here the lower limit a is fixed.

Example. $\frac{d}{dx} \int_1^x \sqrt{1+t^4} dt = \sqrt{1+x^4}$

Q02.19. Find $\frac{d}{dx} \int_1^{\sin x} \sqrt{1+t^4} dt$.

Solution. We apply the FTC Part 1 and the Chain Rule:

Let $u = \sin x$.

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

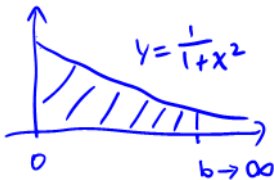
$$\begin{aligned} & \frac{d}{dx} \int_1^{\sin x} \sqrt{1+t^4} dt \\ &= \frac{d}{dx} \int_1^u \sqrt{1+t^4} dt \\ &= \frac{d}{du} \int_1^u \sqrt{1+t^4} dt \cdot \frac{du}{dx} \\ &= \sqrt{1+u^4} \cdot \frac{du}{dx} \\ &= \sqrt{1+(\sin x)^4} \cdot \cos x. \end{aligned}$$

Q03.01. Evaluate the improper integral

$$\int_0^{\infty} \frac{1}{1+x^2} dx. = \lim_{b \rightarrow \infty} \int_0^b \frac{1}{1+x^2} dx$$

Q03.02. Evaluate the improper integral

$$\int_0^1 \frac{1}{(x-1)^{\frac{2}{3}}} dx.$$



Improper Integrals

Improper Integrals refer to those 'definite integrals' where

(i) the interval of integration is not a finite closed interval

$[a, b]$, or

(ii) the integrand is not continuous or not defined at certain point in **$[a, b]$** .

Q03.01. Evaluate the improper integral

$$\int_0^{\infty} \frac{1}{1+x^2} dx.$$

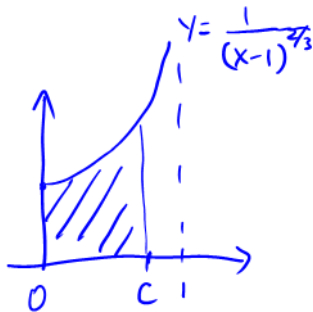
Solution.

$$\begin{aligned} & \int_0^{\infty} \frac{1}{1+x^2} dx \\ &= \lim_{b \rightarrow \infty} \int_0^b \frac{1}{1+x^2} dx \\ &= \lim_{b \rightarrow \infty} \left[\tan^{-1} x \right]_0^b \\ &= \lim_{b \rightarrow \infty} (\tan^{-1} b - \tan^{-1} 0) = \frac{\pi}{2} - 0 = \frac{\pi}{2}. \end{aligned}$$

Q03.02. Evaluate the improper integral

$$\int_0^1 \frac{1}{(x-1)^{\frac{2}{3}}} dx.$$

Solution.



$$\begin{aligned} & \int_0^1 \frac{1}{(x-1)^{\frac{2}{3}}} dx \\ &= \lim_{c \rightarrow 1^-} \int_0^c \frac{1}{(x-1)^{\frac{2}{3}}} dx \\ &= \lim_{c \rightarrow 1^-} \left[3(x-1)^{\frac{1}{3}} \right]_0^c \\ &= \lim_{c \rightarrow 1^-} 3(c-1)^{\frac{1}{3}} - 3(0-1)^{\frac{1}{3}} = 0 - (-3) = 3. \end{aligned}$$

Q03.3. Find the area of the region bounded by the curve $y = \ln(x + 3)$, the curve $y = 2 \ln x$ and the x -axis.

Q03.4. (Disk Method) The region bounded by curve $y^2 = x$, the line $y = x - 2$ and the y -axis is rotated about the x -axis. Find the volume of the solid generated.

Q03.5. (Shell Method) The region bounded by the curve $y = \ln(x - 1)$, the x -axis, the y -axis and the line $y = \ln 2$ is revolved about the x -axis. Use the shell method to find the volume of the solid generated.

Q03.6. Find the length of the curve

$$y = \frac{x^2}{8} - \ln x, \quad 1 \leq x \leq 2.$$

5. Applications of Definite Integrals (Chapter 6)

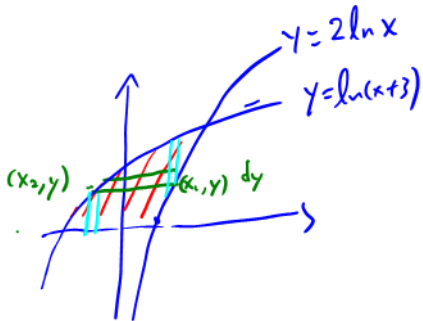
- Area between curves
- Volume of solid of revolution by Disk Method
- Volume of solid of revolution by Cylindrical Shell Method
- Arc length of a curve

Q03.3. Find the area of the region bounded by the curve $y = \ln(x + 3)$, the curve $y = 2 \ln x$ and the x -axis.

Solution:

$$\begin{aligned} \text{Area} &= \int_0^{1.668} (x_1 - x_2) dy \\ &= \int_0^{1.668} (e^{\frac{y}{2}} - (e^y - 3)) dy \end{aligned}$$

$$\begin{aligned} y &= \ln(x_2 + 3) \\ \Rightarrow x_2 &= e^y - 3 \end{aligned}$$



$$\begin{aligned} y &= 2 \ln x_1 \\ \Rightarrow x_1 &= e^{\frac{y}{2}} \end{aligned}$$

Solution: First we find the point of intersection of the two curves:

$$\begin{cases} y = \ln(x + 3) & (1) \\ y = 2 \ln x & (2) \end{cases}$$

From (1) and (2), we have

$$\ln(x + 3) = 2 \ln x = \ln(x^2)$$

$$\implies x^2 - x - 3 = 0 \implies x = \frac{1 \pm \sqrt{13}}{2}.$$

The negative root $\frac{1 - \sqrt{13}}{2}$ is rejected, since

$2 \ln \frac{1 - \sqrt{13}}{2}$ is not defined.

$$\text{When } x = \frac{1 + \sqrt{13}}{2} \approx 2.303,$$

$$y = 2 \ln \frac{1 + \sqrt{13}}{2} \approx 1.668.$$

Thus the intersection point is **(2.303, 1.668)**.

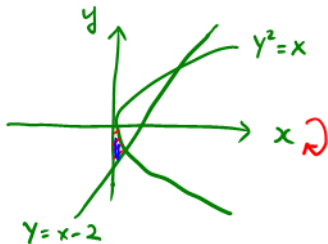
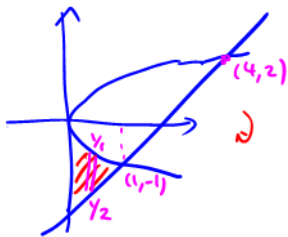
Now

$$y = \ln(x + 3) \iff e^y = x + 3 \iff x = e^y - 3,$$

$$y = 2 \ln x \iff \frac{y}{2} = \ln x \iff x = e^{y/2}.$$

Then

$$\begin{aligned}\text{Area} &= \int_0^{1.668} (\mathbf{e^{y/2}} - (\mathbf{e^y} - \mathbf{3})) \, dy \\ &= \left[\mathbf{2e^{y/2}} - \mathbf{e^y} + \mathbf{3y} \right]_0^{1.668} \\ &= (\mathbf{2e^{1.668/2}} - \mathbf{e^{1.668}} + \mathbf{3(1.668)}) - (\mathbf{2} - \mathbf{1} + \mathbf{0}) \\ &\approx \mathbf{3.31}.\end{aligned}$$



Q03.4. (Disk Method) The region bounded by curve $y^2 = x$, the line $y = x - 2$ and the y -axis is rotated about the x -axis. Find the volume of the solid generated.

$$\text{Volume} = \int_0^1 \pi (y_2^2 - y_1^2) dx = \int_0^1 \pi ((x-2)^2 - (-\sqrt{x})^2) dx$$

$$y_2 = x - 2$$

$$y_1^2 = x \\ \Rightarrow y_1 = \sqrt{x} \text{ or } -\sqrt{x}$$

Solution: First we find the intersection points of the parabola and the line:

$$\begin{cases} y^2 = x & (1) \\ y = x - 2 & (2) \end{cases}$$

Substituting (2) into (1), we have

$$\begin{aligned} (x - 2)^2 = x &\implies x^2 - 5x + 4 = 0 \\ &\implies (x - 1)(x - 4) = 0 \implies x = 1, 4. \end{aligned}$$

When $x = 1$, $y = 1 - 2 = -1$.

When $x = 4$, $y = 4 - 2 = 2$.

Thus the points of intersection are $(1, -1)$ and $(4, 2)$.

For the lower half of the parabola, we have

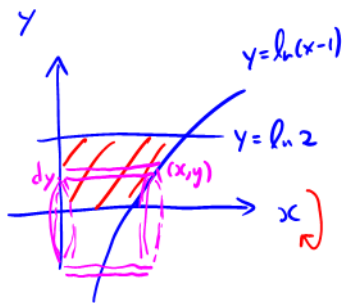
$$y^2 = x \implies y = -\sqrt{x}.$$

Then

$$\begin{aligned}\text{Volume} &= \int_0^1 \pi((x-2)^2 - (-\sqrt{x})^2) dx \\&= \pi \int_0^1 ((x-2)^2 - x) dx \\&= \pi \cdot \left[\frac{(x-2)^3}{3} - \frac{x^2}{2} \right]_0^1 \\&= \pi \left(\left(\frac{(1-2)^3}{3} - \frac{1^2}{2} \right) - \left(\frac{(0-2)^3}{3} - 0 \right) \right) \\&= \frac{11}{6} \pi.\end{aligned}$$

Q03.5. (Shell Method) The region bounded by the curve $y = \ln(x - 1)$, the x -axis, the y -axis and the line $y = \ln 2$ is revolved about the x -axis. Use the shell method to find the volume of the solid generated.

$$\begin{aligned} \text{Volume} &= \int_0^{\ln 2} 2\pi y \cdot x \, dy \\ &= \int_0^{\ln 2} 2\pi \cdot y \cdot (e^y + 1) \, dy \end{aligned}$$



$$y = \ln(x-1) \Rightarrow x = e^y + 1$$

Solution: Note that $y = \ln(x - 1) \implies x = e^y + 1$.

Then

$$\begin{aligned}\text{Volume} &= \int_0^{\ln 2} 2\pi y \cdot x \, dy \\&= \int_0^{\ln 2} 2\pi y \cdot (e^y + 1) \, dy \\&= 2\pi \left(\left[y \cdot (e^y + y) \right]_0^{\ln 2} - \int_0^{\ln 2} 1 \cdot (e^y + y) \, dy \right) \\&= 2\pi \left(\left[y \cdot (e^y + y) \right]_0^{\ln 2} - \left[e^y + \frac{y^2}{2} \right]_0^{\ln 2} \right) \\&= 2\pi \cdot \left[\frac{y^2}{2} + (y - 1)e^y \right]_0^{\ln 2} \\&= 2\pi \left(\frac{(\ln 2)^2}{2} + (\ln 2 - 1)e^{\ln 2} - (0 - 1) \right) \\&= \pi((\ln 2)^2 + 4 \ln 2 - 2) \approx 3.94.\end{aligned}$$

Q03.6. Find the length of the curve

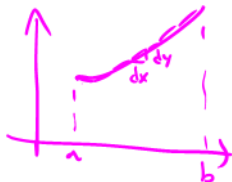
$$y = \frac{x^2}{8} - \ln x, \quad 1 \leq x \leq 2.$$

Recall: For a curve $y = f(x)$, $a \leq x \leq b$,

$$\text{Length of curve} = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

(To visualize the formula, think of the curve as consisting of many small triangles of width dx and height dy , so that the hypotenuse is of length $\sqrt{(dx)^2 + (dy)^2}$).

$$\begin{aligned} \text{length} &= \int_a^b \sqrt{(dx)^2 + (dy)^2} \\ &= \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \end{aligned}$$



Solution: Note that

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{x^2}{8} - \ln x \right) = \frac{x}{4} - \frac{1}{x}.$$

Then

Length of curve

$$\begin{aligned} &= \int_1^2 \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx \\ &= \int_1^2 \sqrt{1 + \left(\frac{x}{4} - \frac{1}{x} \right)^2} dx \end{aligned}$$

Length of curve

$$= \int_1^2 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$= \int_1^2 \sqrt{1 + \left(\frac{x}{4} - \frac{1}{x}\right)^2} dx$$

$$= \int_1^2 \sqrt{1 + \frac{x^2}{16} - 2 \cdot \frac{x}{4} \cdot \frac{1}{x} + \frac{1}{x^2}} dx$$

$$= \int_1^2 \sqrt{\frac{x^2}{16} + \frac{1}{2} + \frac{1}{x^2}} dx$$

$$= \int_1^2 \sqrt{\left(\frac{x}{4} + \frac{1}{x}\right)^2} dx = \int_1^2 \left(\frac{x}{4} + \frac{1}{x}\right) dx$$

$$= \left[\frac{x^2}{8} + \ln|x|\right]_1^2 = \left(\frac{2^2}{8} + \ln 2\right) - \left(\frac{1^2}{8} + \ln 1\right) = \frac{3}{8} + \ln 2.$$